

CODECOMPASS EVALUATION

CodeCompass Grounded QA Evaluation Synthesis

Presentation-oriented summary of two frozen bilingual repository evaluations

Evidence supports READY_WITH_LIMITATIONS for GLM-backed QA

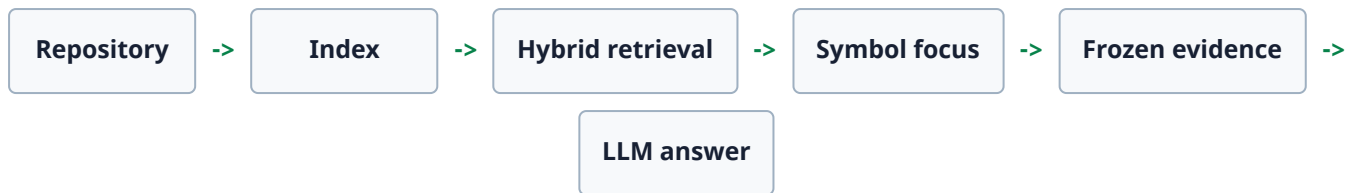
Hospital-System: GLM full matrix, 8 questions
CS-Bookstore: controlled Qwen-vs-GLM matrix, 12 questions
No cross-protocol pooled accuracy

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1. Evaluation objective and architecture

The evaluation asks whether CodeCompass can retrieve source-grounded evidence, preserve trusted citation identity, and generate concise bilingual answers without altering source metadata.



Trusted citation path: SQLite project/file/symbol/chunk metadata supplies relative paths and line ranges independently of LLM prose.

Interpretation separates retrieval correctness, generation correctness, citation validity, and language adherence.

2. Repositories and experiment designs

Dimension	Hospital-System	CS-Bookstore
Project	2 / Hospital-System	3 / CS-Bookstore
Repository pin	Not recorded in authoritative artifact	56ee148f3b4b8bc1f4eaa921e22f280fa14ad7a
Files	33	56
Symbols	409	144
Chunks / vectors	409 / 409	144 / 144
Frozen questions	8	12
Model coverage	GLM full matrix only	Qwen and GLM on identical prompt/context pairs
Valid quantitative model comparison	No full-matrix local-vs-cloud comparison	Yes, within this 12-question matrix

Do not pool the result rates. Hospital-System and CS-Bookstore use different model coverage and protocols. No combined overall model accuracy is reported.

Hospital [HS-QA-01](#) is a known regression anchor; the remaining seven Hospital questions are novel frozen questions. The authoritative Hospital artifact records a successful production-path smoke but does not retain enough local-Qwen A/B detail for a quantitative comparison.

3. Observed results

Hospital-System full matrix - GLM only

7

1

PASS
7/8

FAIL
1/8

VERDICT
READY_WITH_LIMITATIONS

CS-Bookstore controlled full comparison

Metric	Local Qwen	GLM
PASS / PARTIAL / FAIL / INCONCLUSIVE	1 / 4 / 6 / 1	10 / 2 / 0 / 0
English PASS	1/6	6/6
Persian PASS	0/6	4/6
Unsupported claims	7	0
Missing facts	24	3
Verdict	NOT_READY	READY_WITH_LIMITATIONS

4. Retrieval, generation, citations, and language

Layer	Hospital-System observation	CS-Bookstore observation
Retrieval	No failure identified in the recorded matrix summary	Expected target retrieved and retained 12/12; retrieval failures 0
Generation	7 PASS, 1 unsupported-type FAIL	Qwen: 10 generation-attributed non-PASS plus 1 inconclusive; GLM: 2 omission-only non-PASS
Citations	13/13 valid	Qwen 14/14; GLM 14/14
Language adherence	8/8	12/12 for each model

Citation validity remained intact despite generation errors. This demonstrates separation between model prose quality and trusted navigation metadata. It does not imply that every retrieved context set is optimally selective.

Observed Persian and English counts are descriptive only. Neither sample supports a broad language-quality claim.

5. Failure examples

Hospital-System: unsupported type claim

The requested helper-object facts were complete, but the unsupported parameter-type claim is a material generation/prose grounding failure under the frozen rubric.

The requested helper-object facts were correct, but the added string-parameter claim was not source-supported. Citation validity remained intact.

CS-Bookstore: unsupported uniqueness claim

The conditional slug generation was described, but the parent save call was omitted and uniqueness was invented.

CS-Bookstore: omission on shared evidence

For [CSB-QA-08](#), GLM correctly described search construction but omitted the request.GET/form-validation step. Qwen's response ended mid-sentence and was recorded as INCONCLUSIVE because finish_reason was not retained.

These examples show why retrieval presence, complete generation, and citation validity must be scored separately.

6. Latency and local-vs-cloud trade-off

Experiment/model	Mean	Median	Range
Hospital-System / GLM	8.671s	8.562s	5.494-12.609s
CS-Bookstore / local Qwen	24.277s	20.373s	7.948-54.785s
CS-Bookstore / GLM	13.547s	11.717s	8.500-31.303s

Latency values are descriptive measurements from these runs, not deployment guarantees. The Qwen/GLM token budgets differ by design, so the comparison is not equal-token.

Local Qwen strengths	Cloud GLM strengths observed here
Offline execution; local privacy; no cloud-generation cost	More complete grounded answers on the controlled CS-Bookstore matrix; lower observed mean and median latency in that run

Provider usage and monetary cost were not retained. No cost estimate is invented.

7. Limitations and overall assessment

- Two repositories and small frozen matrices do not support statistical significance or universal model claims.
- Hospital-System did not run a full local-Qwen matrix, so only CS-Bookstore supports quantitative local-vs-cloud comparison.
- One generation per model/question does not measure run-to-run variance.
- Token budgets differ between Qwen and GLM to avoid artificial truncation.
- Usage, reasoning tokens, finish_reason, and monetary cost were not retained by the provider response abstraction.
- Valid citations prove trusted source navigation, not complete or error-free prose.

Overall technical assessment

The frozen evidence supports CodeCompass as a citation-preserving grounded-QA system with GLM-backed generation marked `READY_WITH_LIMITATIONS`. The local Qwen configuration is `NOT_READY` for the strict CS-Bookstore QA workload, while retaining offline and privacy advantages. No claim of hallucination-free behavior or general retrieval perfection is made.

8. Reproducibility summary

Artifact	Identity
Hospital matrix	3fbeac128da1251ebc3cd262abcb3208044f058a53e3b94977c5e4fddebbe44d
Hospital rubric	17cc25b4435b9a44189c2ff87b7e04b800fb1a74a8fbd5de523ef7e53151638e
CS-Bookstore matrix	72c371f058a9301f034d291240f60545d3b3c85d5ae8fb9446a66fab24ee5252
CS-Bookstore rubric	f1a2af66613d1ee2713a1de6dc699603add0e5c4132c3b6875ae4d35e28234f7
CS-Bookstore repository commit	56ee148f3b4b8bc1f4eaa921e22f280fa14ad7a

All values in this synthesis are projections of the frozen JSON/CSV artifacts. No new LLM request, retrieval, benchmark run, or metric reconstruction was performed.